FIFO
8 × 8

Status FIFO

Synchronous FIFO with almost flags and rejected-operation counters

Single clock | Almost-full and almost-empty flags | Saturating reject counters

Verilog HDL

LEVEL 3

Definition & Purpose

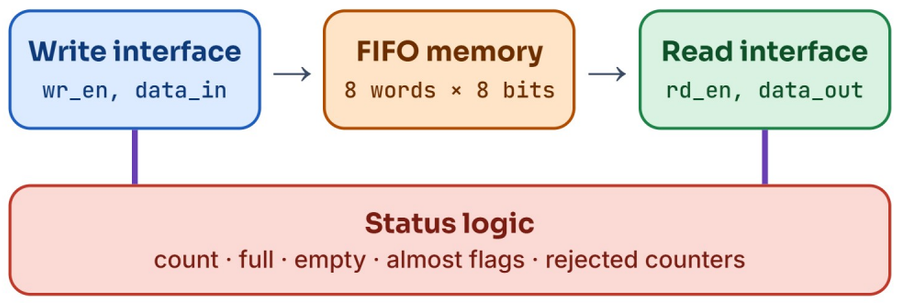
A synchronous FIFO is a First-In-First-Out buffer driven by a **single clock**. This version adds early-warning flags and counts every request it had to turn away.

- ✓ Almost-full and almost-empty thresholds
- ✓ Counts writes rejected when full and reads rejected when empty
- ✓ Parameterized width, depth and thresholds

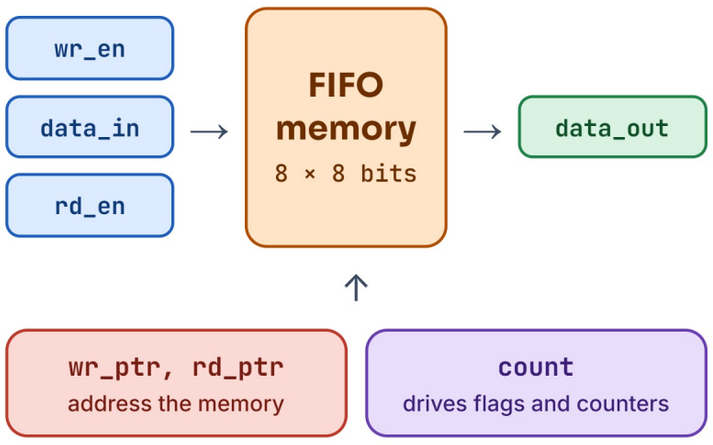
Parameters

`DATA_WIDTH`	**8**	bits per word
`DEPTH`	**8**	words stored
`ADDR_WIDTH`	**3**	log2 of 8
`COUNT_WIDTH`	**4**	holds 0 to 8
`ALMOST_FULL_LEVEL`	**6**	flag at count ≥ 6
`ALMOST_EMPTY_LEVEL`	**2**	flag at count ≤ 2
`ERROR_WIDTH`	**8**	counters stop at 255

Block Diagram

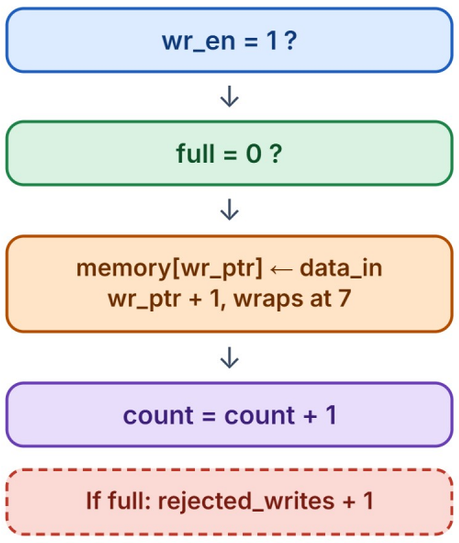


Internal Components

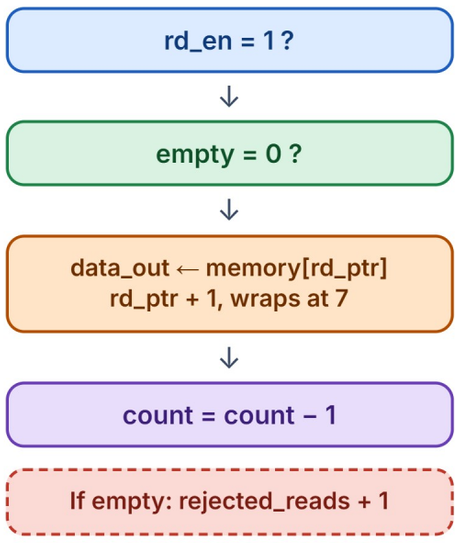


Reset clears both pointers, **count**, **data_out** and both reject counters. The memory array itself is not cleared.

Write Flow



Read Flow



Flag Logic

```
full = (count == 8)
empty = (count == 0)
almost_full = (count ⩾ 6)
almost_empty = (count ⩽ 2)
```

Default parameters. Almost flags are also true when the FIFO is full or empty.

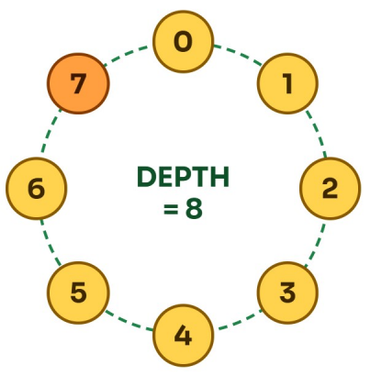
Count Update

Accepted	Count change
Write only	**+1**
Read only	**−1**
Both	**No change**
Neither	**No change**

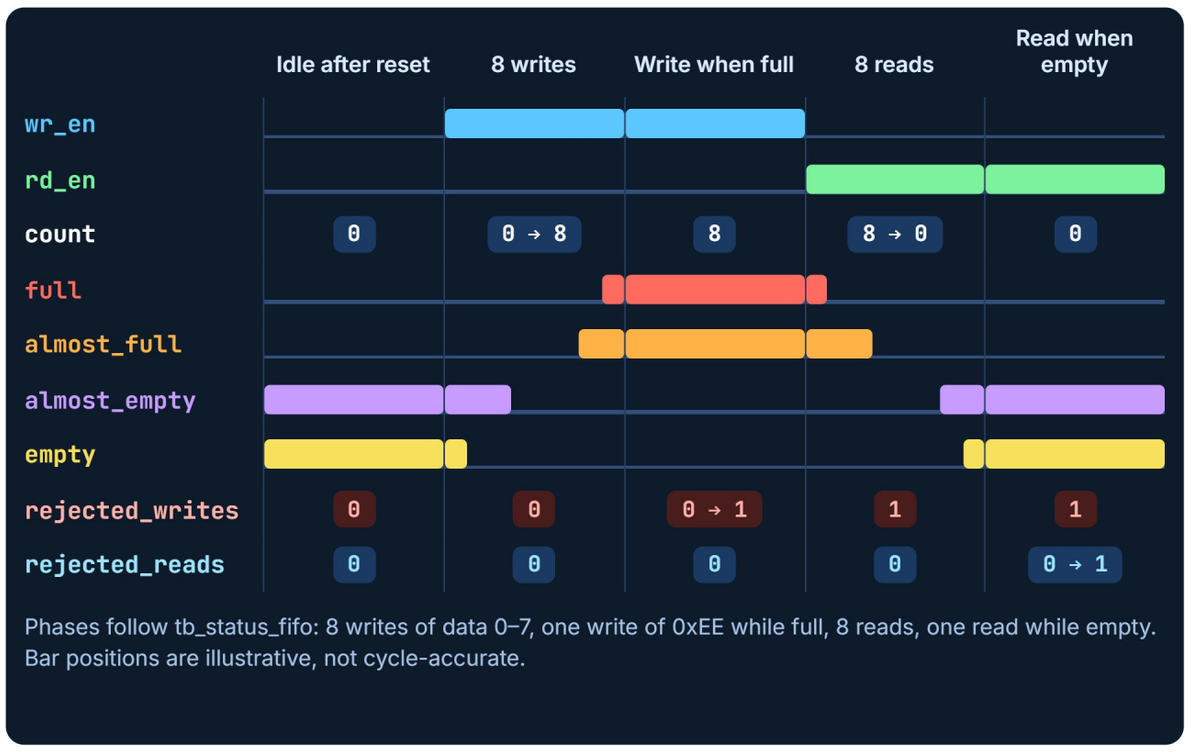
Pointer Wraparound

A pointer that reaches DEPTH − 1 (7) wraps back to 0, so 8 addresses are reused in a ring.

```
if (wr_ptr == DEPTH-1) wr_ptr ⩽ 0;
```

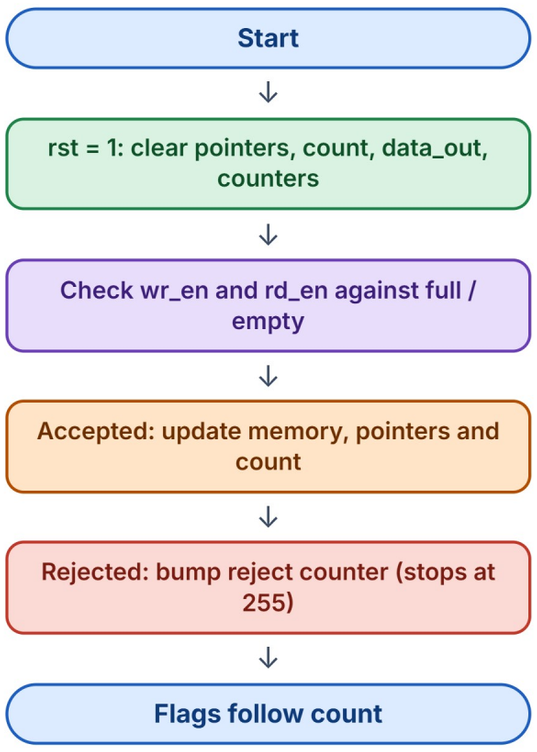


Testbench Timeline



Phases follow tb_status_fifo: 8 writes of data 0–7, one write of 0xEE while full, 8 reads, one read while empty. Bar positions are illustrative, not cycle-accurate.

Code Flow



Verification

tb_status_fifo checks, in order:

- ✓ 8 writes: full, almost_full, count = 8
- ✓ Extra write is counted in rejected_writes
- ✓ 8 reads: empty and almost_empty are set
- ✓ Extra read is counted in rejected_reads

LEVEL 3 PASS

Printed by the testbench when every check succeeds

Rejected-Operation Counters

```
rejected_writes + 1 when
wr_en && full && !(rd_en && !empty)
```

```
rejected_reads + 1 when
rd_en && empty && !(wr_en && !full)
```

1. A write is not counted if a read frees a slot in the same cycle, and the reverse for reads.
2. Saturating: at 8'hFF (255) a counter holds instead of wrapping to 0.

Key Takeaways

- ✓ Almost flags warn at 6 and 2, before full or empty
- ✓ Rejected requests are counted, never silently lost
- ✓ Change DEPTH together with ADDR_WIDTH and COUNT_WIDTH

Conclusion

status_fifo extends a basic synchronous FIFO with threshold flags and saturating counters for rejected writes and reads. A controlling design can slow down early and see how many requests were turned away.